



Sunspots, Volcanic Eruptions and Earthquakes in R: Periodicities and Correlations

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Introduction:

Sunspots

- Representative of solar magnetic activity
- Known to fluctuate on an 11 year cycle
- Other, weaker periodicities appear to exist

Volcanoes

- The specific factors that cause volcanoes to erupt are unclear
- Stothers 1989 suggests that there may be a correlation between solar and volcanic activity
 - Corroborated by my research under Dr. Marsellos last year
- Importance: much of the world is affected by the possibility of volcanic eruptions, so it is necessary to improve eruption prediction methods.

Earthquakes

- Stothers 1989 proposed that solar activity triggers small earthquakes, releasing the pressure and decreasing the frequency of eruptions
 - This research tests that claim

Materials:

Data

- Sunspot Index and Long-term Solar Observations (SILSO) provided sunspot numbers
- National Centers for Environmental Information (NCEI) provided eruption data
- U.S. Geological Survey provided earthquake data

Software

- All code was run in RStudio
- All graphs were produced with RStudio
- Packages used: dplyr, tidyverse, WaveletComp, scorepeak, kza, data.table, lubridate, httr, jsonlite, maps, BBmisc, knitr, zoo, TSA

Methods:

- Obtained earthquake, sunspot, and eruption data
- Confirmed a correlation between solar cycle and eruptions using a periodogram
- Added a cutoff date to both USGS and SILSO data to prevent changes in measuring techniques from affecting the results
- Detrended both data sets
- Ran a cross correlation function (CCF) and obtained ambiguous results
- Applied a prewhitening filter based on the sunspot data to both data sets; ran CCF again

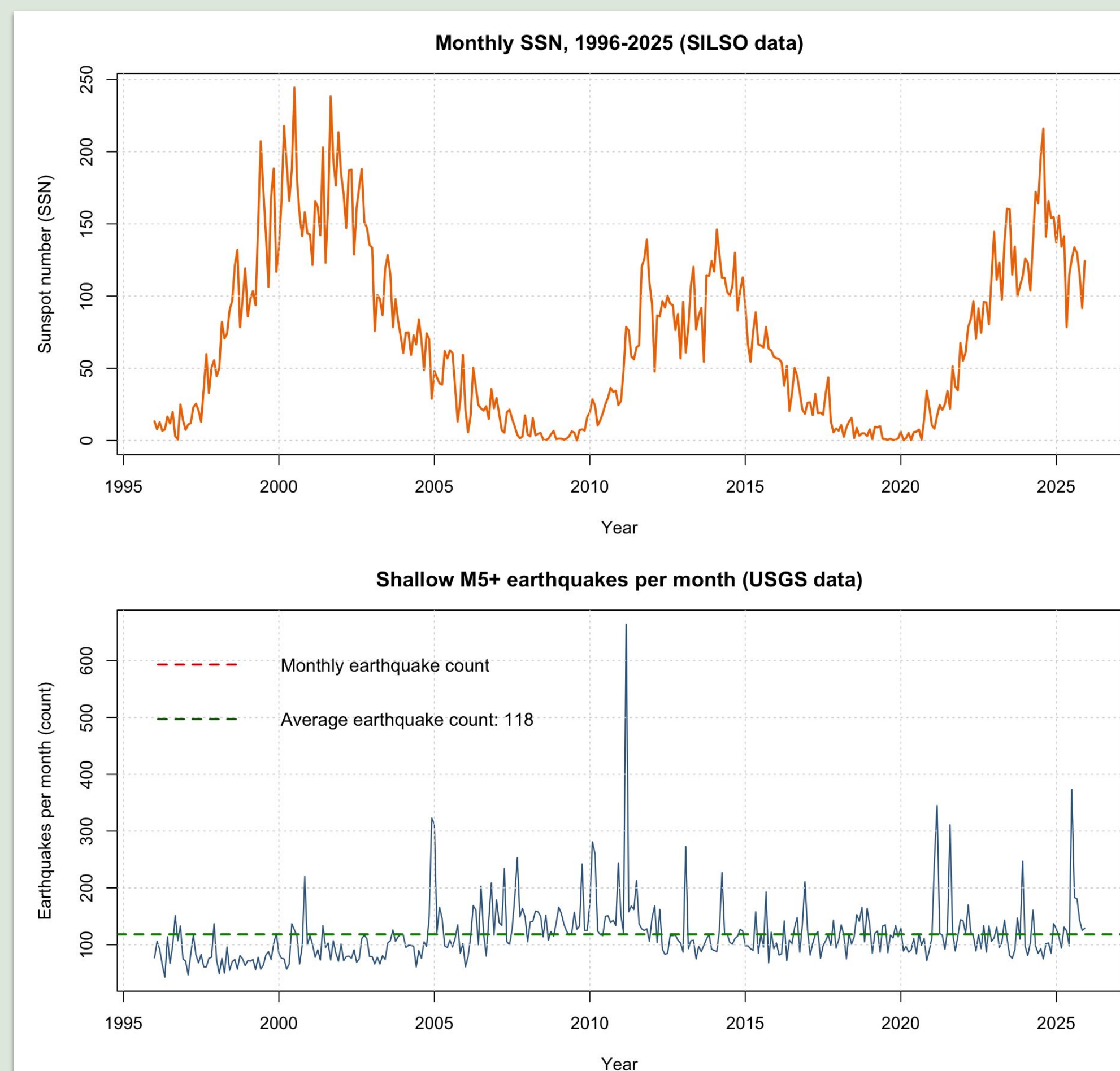


Figure 1a: Graph showing monthly SSN in the study window. The analysis begins in 1996 due to improved measurement techniques. **Figure 1b:** Graph showing monthly earthquake counts, as well as the average earthquake count in the study window. The analysis begins in 1996 due to improved measurement techniques.

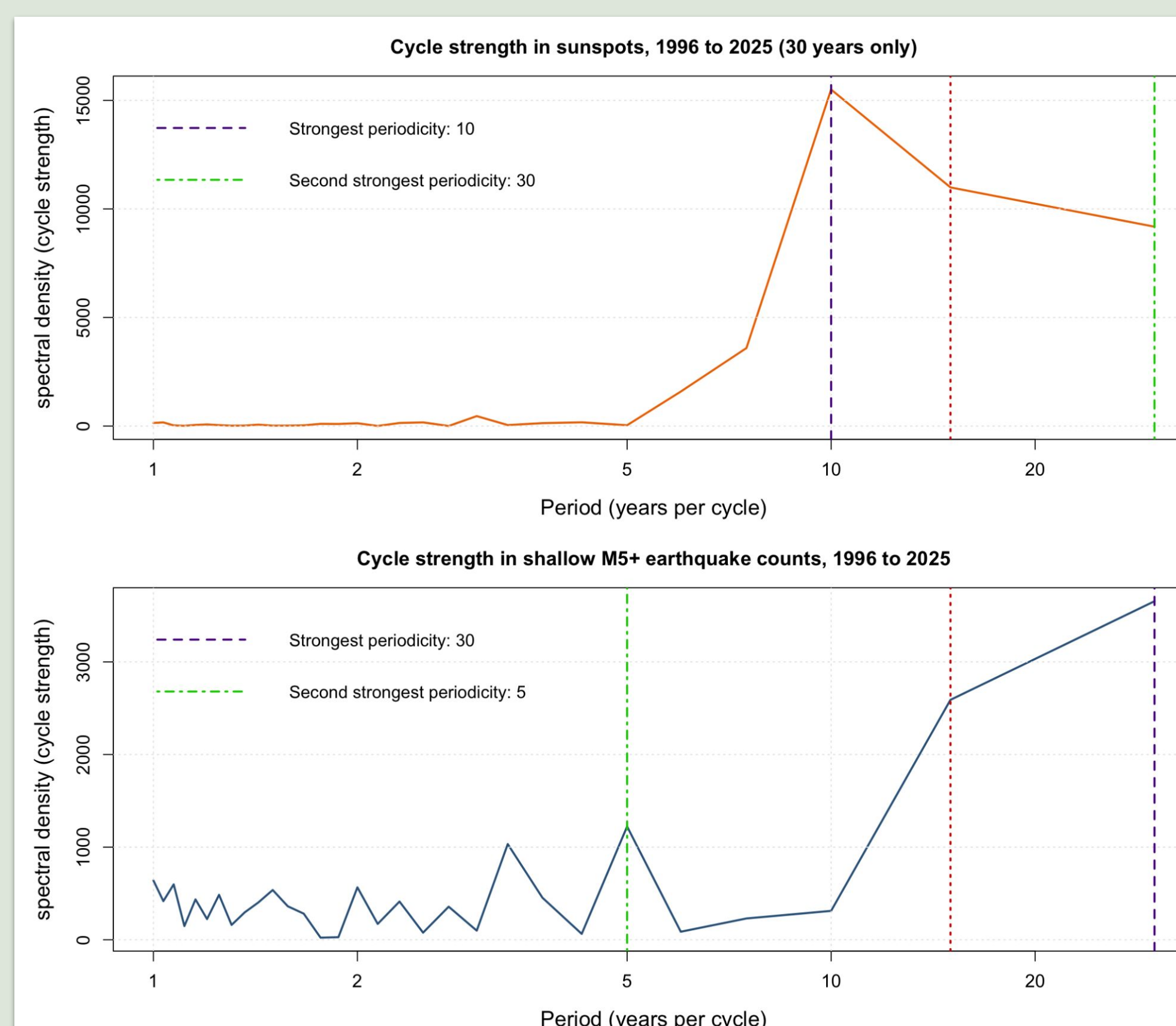


Figure 2a: Graph showing the strength of sunspot cycles over a 30 year period. The peak is at 10 years due to the software's closest matching periodicity being $30/3=10$ years, indicating that the periodogram is not a useful tool at this small of a year range. **Figure 2b:** Graph showing the strength of earthquake cycles over a 30 year period. The peak is at 30 years-the length of the entire record, and therefore useless information.

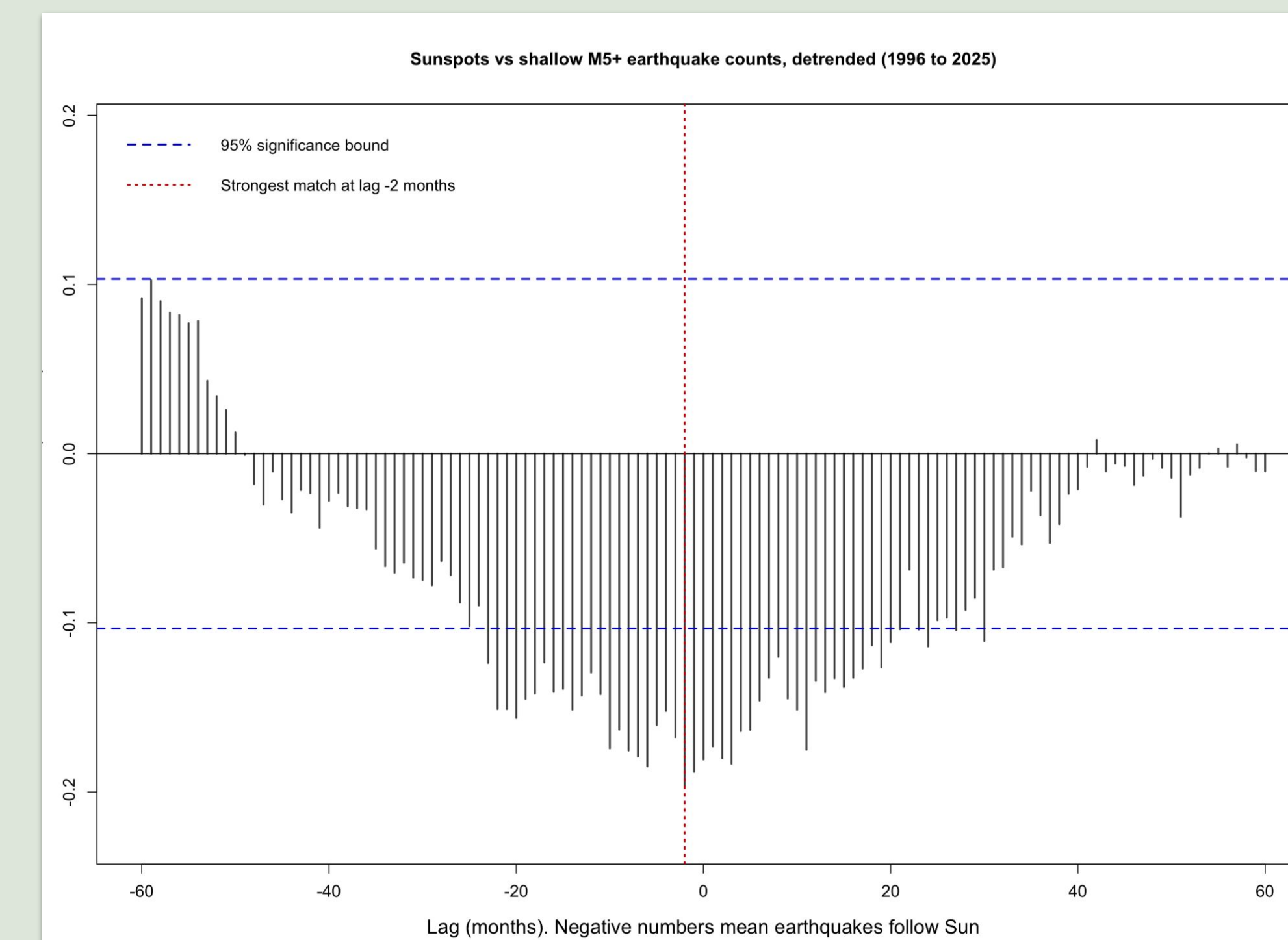


Figure 3: Graph showing the results of a CCF applied to the sunspot and earthquake data. While it is tempting to identify the strongest lag (dotted red line) as important, the shape of the ccf and the fact that 49 bars cross the 95% significance line (dashed blue line) suggest that this is possibly a product of both data sets being smooth series.

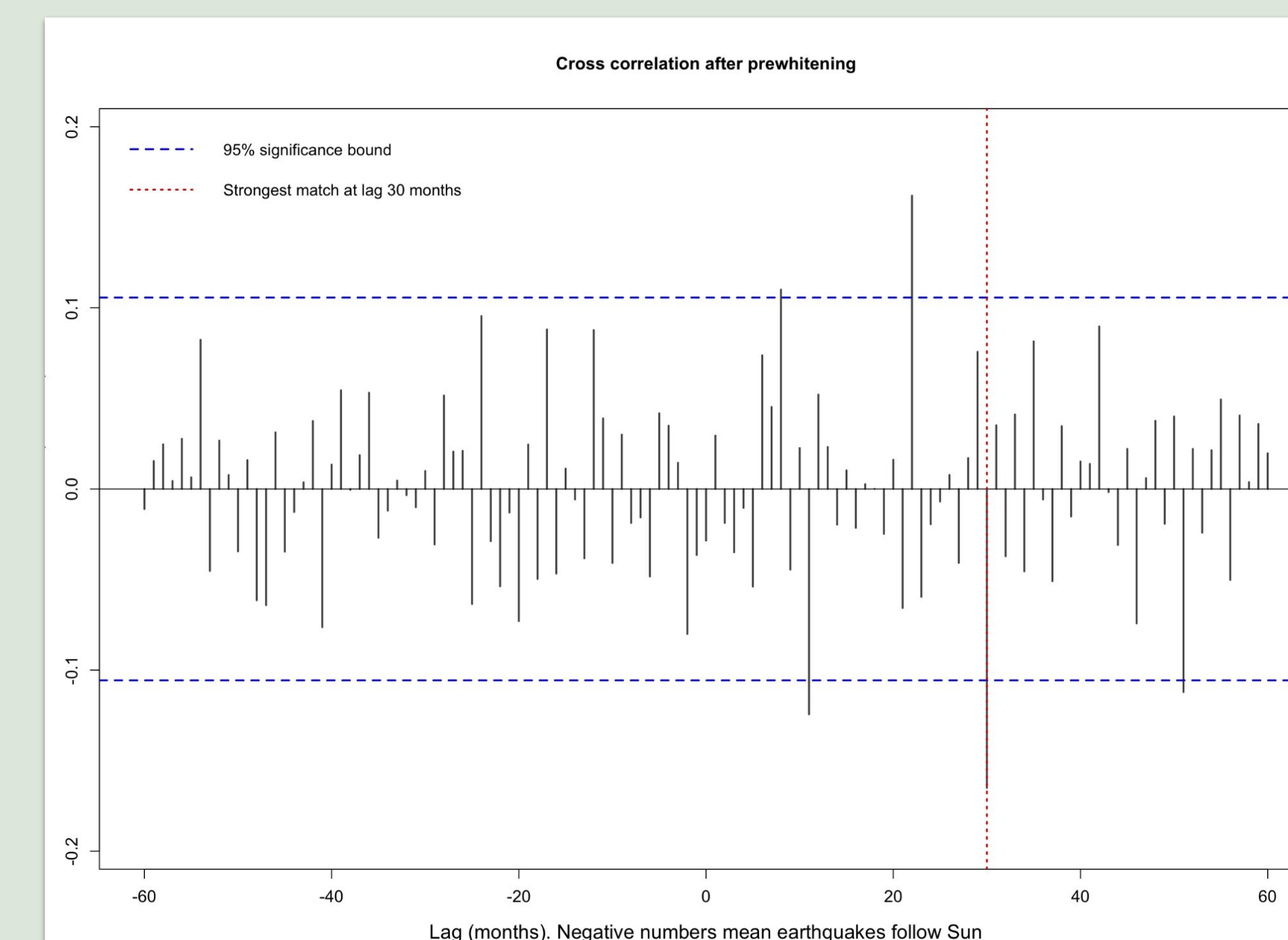


Figure 4: Graph showing the results of a CCF after prewhitening was applied to both the sunspot and earthquake data. Using R's autoregressive model applied to the sunspot data, a lag order of 16 months was selected with AIC. After prewhitening, only 5/121 bars cross the significance bound line (dashed blue lines), which is less than the 6/121 expected bars due to chance. Also, the most prominent lag is at 30 months; a physical impossibility, as earthquakes do not cause solar activity.

Future Research:

- Test Stothers 1989's other hypothesis: correlations between the solar cycle and Earth's climate

Results:

- Some matching periodicities when a periodogram was applied to sunspot data and eruption data (see sunspot_periodogram.csv and eruption_periodogram.csv in GitHub), indicating a possible correlation
- The periodogram was an ineffective tool given the 30 year time period of analysis for sunspots/earthquakes
- 13.1 earthquakes/decade trend was observed after 1996; this was likely due to improved measurements and was removed by detrending
- Largest correlation lag (before prewhitening): -2 months
- Largest correlation lag (after prewhitening): 30 months (noise)
- Number of bars passing the significance bound (before prewhitening): 49
- Number of bars passing the significance bound (after prewhitening): 5
- Number of bars that are expected to pass the significance bound due to chance: around 6
- Conclusion: there is no correlation between solar activity and earthquakes

References:

- Hathaway, D.H. The Solar Cycle. *Living Rev. Sol. Phys.* 12, 4 (2015).
<https://doi.org/10.1007/lrsp-2015-4>
- Stothers, R.B., 1989: Volcanic eruptions and solar activity. *J. Geophys. Res.*, 94, 17371-17381, doi:10.1029/JB094iB12p17371.
- National Geophysical Data Center / World Data Service (NGDC/WDS):
NCEI/WDS Global Significant Volcanic Eruptions Database.
NOAA National Centers for Environmental Information.
doi:10.7289/V5JW8BSH Accessed 30 Jul. 2025.
- WDC-SILSO, Royal Observatory of Belgium, Brussels,
<https://doi.org/10.24414/qnza-ac80>.

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